

IDTA Coursework 1



School of Computing

M33147-INTELLIGENT DATA AND TEXT ANALYTICS

Coursework: item 1

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Module Lecturers: Dr Atefeh Khazaei, Dr Ella Haig, Dr Alaa Mohasseb,
Dr Grace Golcarenenrenji

Student ID(s): up2102739 & up2110125

1.Descriptive analytics

This data set highlights the relation between various professional demographics, and both work related and lifestyle factors. This investigation aims to provide insight into how some factors may impact others, and which attributes are most influential. The dataset will be explored using a range of exploratory data analysis, predictive modelling, plus statistical research. Ultimately using lifestyle patterns to predict mental health outcomes.

1.1 Statistics For All Attributes

With 2054 professional entries, this dataset has 11 attributes. Five which are numerical and six that are categorical.

Table 1.1 Attributes and their datatypes.

Numerical	Categorical
age	gender
work pressure	sleep duration
job satisfaction	dietary habits
work hours	suicidal thoughts
financial stress	family history
	depression

Table 1.2 Numerical attributes and their statistical analysis

	Age	Work Pressure	Job Satisfaction	Work Hours	Financial Stress
count	2054.000000	2054.000000	2054.000000	2054.000000	2054.000000
mean	42.171860	3.021908	3.015093	5.930867	2.978578
std	11.461202	1.417312	1.418432	3.773945	1.413362
min	18.000000	1.000000	1.000000	0.000000	1.000000
25%	35.000000	2.000000	2.000000	3.000000	2.000000
50%	43.000000	3.000000	3.000000	6.000000	3.000000
75%	51.750000	4.000000	4.000000	9.000000	4.000000
max	60.000000	5.000000	5.000000	12.000000	5.000000

This Data set contains individuals 18 to 60 years old. The mean age and median are so close it suggests that the distribution is roughly symmetrical with no strong skew. With standard deviation(std) being 11.46 years, this indicates a broad representation of different aged individuals.

Work pressure has almost an identical mean and median, so there is a fairly symmetric distribution. The std is 1.4 showing considerable variation, covering the full range from minimal to maximum pressure. The interquartile range(IQR) is 2, so 50% of employees have a low to moderate and moderate to high work pressure. However, 75% (Q3) experience pressure level ≤ 4 , which is a large share feeling moderate to high pressure.

Job satisfaction's central tendency indicates a balanced symmetric distribution as the mean and median are almost equal. The average job satisfaction level is moderate. A std of 1.418 shows employees differ noticeably in how satisfied they are. The IQR reflects that 50% of employees have a moderately low to moderately high satisfaction but, $\frac{3}{4}$ report a satisfaction of 4 or below and only the top 25% feel highly satisfied. This wide range may indicate the workforce could be disengaged while some are satisfied.

Work hours varied vastly with a high std of 3.77. The distribution is wide but reasonably balanced, since the mean $\approx$ median. Seeing a range from 0 to 12 hours/day may become a significant independent variable throughout further comparative analysis.

Most employees experience moderate financial stress, the median is 3 and there is no extreme skew. 25% experience high stress (4-5), and 25% experience lower financial stress, showing a broad variability. Overall entries with moderate stress is 50%, so it appears that financial stress is unevenly distributed.

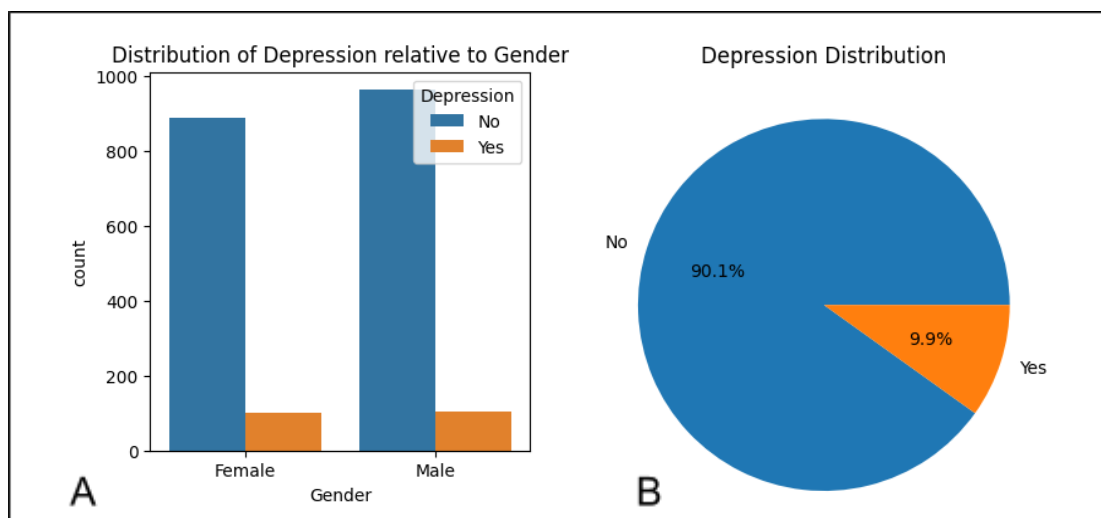
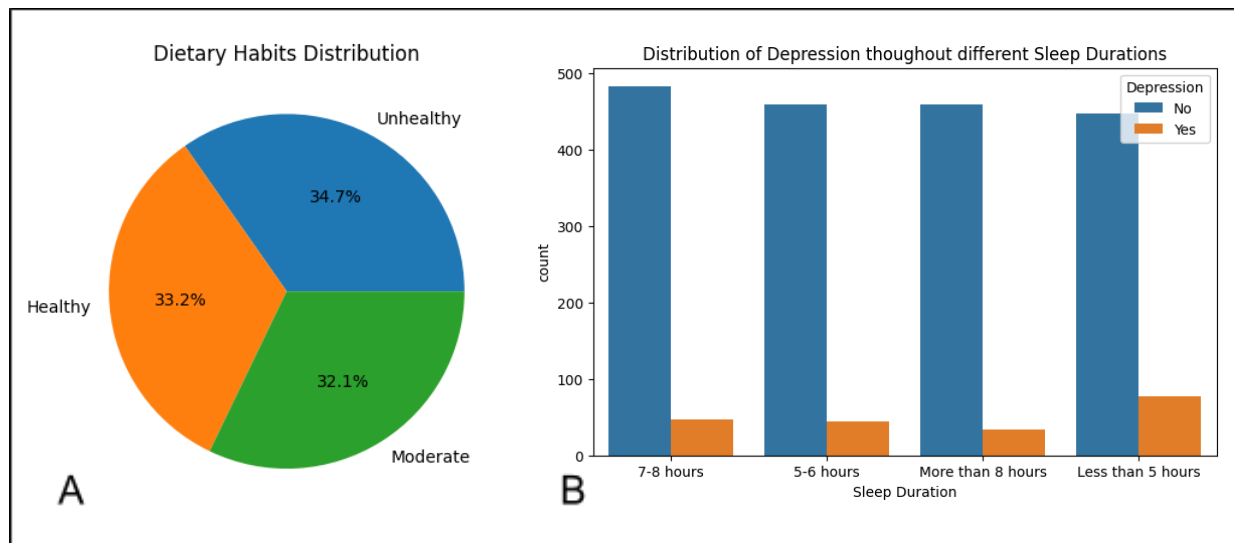


Fig. 1.1 figures showing the distribution of categorical variables. A) histogram of the gender split. B) piechart showing the distribution of depression.

As shown on figure 1.1a there is a very even distribution between male (51.8% [1.dp](#)) and female (48.1% [1.dp](#)) candidates in this dataset with only slightly more men. This gives a good representation from each gender. Figure B illustrates that only 9.9% of entries suffered from depression.



1.3 figures showing the distribution of categorical variables. A) piechart highlighting the dietary habits distribution. B) histogram showing the distribution of sleep duration.

Dietary habits and sleep duration have a very even distribution as shown in figure 1.3. 'Have you ever had suicidal thoughts ?' and 'Family History of Mental Illness' also shares this very balanced distribution shown in table 1.3., providing.

Table 1.3 distribution of Have you ever had suicidal thoughts ? & Family History of Mental Illness

	Have you ever had suicidal thoughts?(3.sf)	Family History of Mental Illness(3.sf)
No	51.8%	50.9%
Yes	48.1%	49.0%

1.2 Data Visualisation

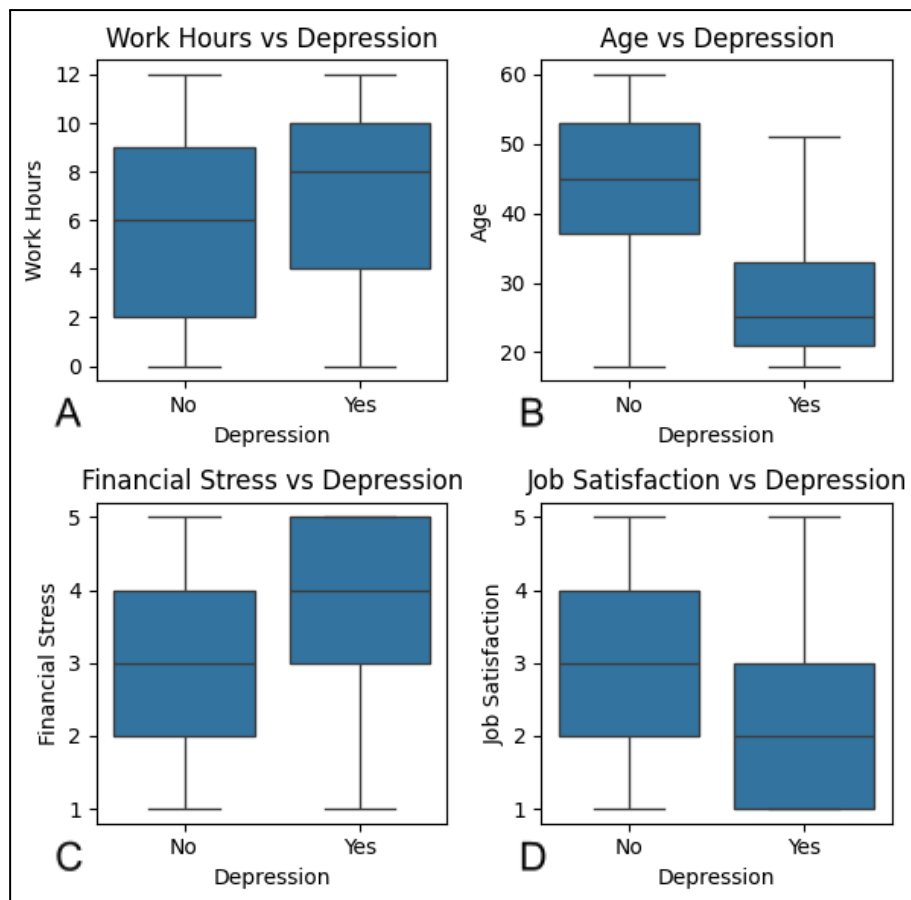


Fig. 1.4 eight box plots comparing four different attributes and their Depression outcome.

Table 1.4 Summary of Variable Comparisons with Depression from Fig.1.4

Comparison	Key Observation	Interpretation
Work Hours vs Depression(Fig.1.4a)	- Individuals with depression show higher median work hours. - IQRs overlap substantially. - Both groups show a wide spread, indicating high variability.	Although depressed individuals appear to work more on average, the overlap in distributions suggests that work hours alone are not a strong predictor of depression
Age vs Depression(Fig.1.4b)	- Non-depressed individuals have a higher median age. - Age range is wider in the non-depressed group. - Younger individuals appear more represented in the depressed	Older individuals seem less likely to be depressed. Younger age groups may report depression more, possibly due to greater awareness and openness about mental health.
Financial Stress vs Depression (Fig.1.4c)	-Depressed individuals experience higher median financial stress. - The upper range extends further for the depressed group. - Distribution suggests stress levels differ strongly between groups	Financial stress shows a strong association with depression and may act as a meaningful independent variable.

Job Satisfaction vs Depression (Fig.1.4d)	- Depressed individuals have notably lower median job satisfaction. - Their range is narrower, indicating consistently low satisfaction. - Non-depressed individuals show higher and more varied satisfaction levels .	Low job satisfaction appears highly linked to depression, potentially acting as both a symptom and a contributing factor.
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Overall fig. 1.4 indicates that the strongest associations with depression are 'Financial Stress' and 'Job Satisfaction' as they show the biggest difference between the groups. 'Work Hours' and 'Age' have weaker associations with more overlap; less direct influence.

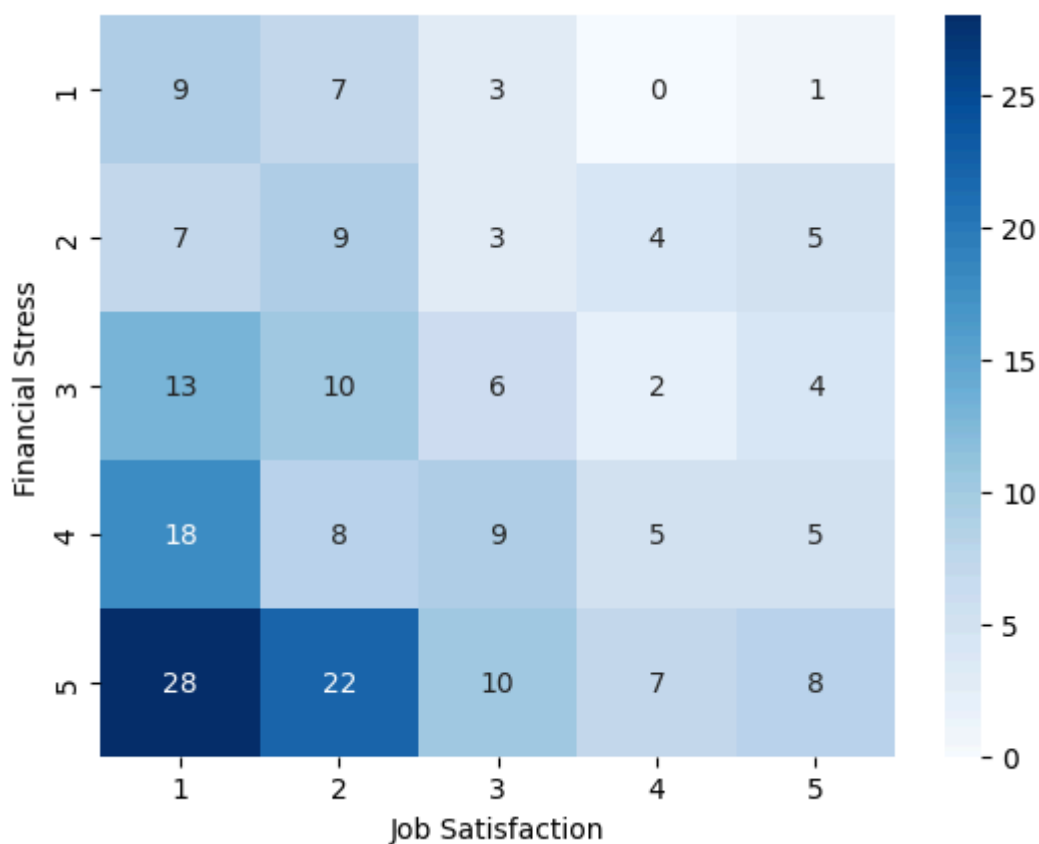


Fig. 1.5 a heatmap- contingency table of two ordinal variables: financial stress and Job Satisfaction

Fig.1.5 shows a strong negative correlation between 'Financial Stress' and 'Job Satisfaction'. The highest frequencies are found with the highest financial stress and lowest job satisfaction. Lowest frequencies are found in the most desirable quadrant (high satisfaction around 4 to 5, and low stress around 1 to 3).

Ultimately this dataset is skewed towards having high stress and low job satisfaction with the opposite being a rare outcome.

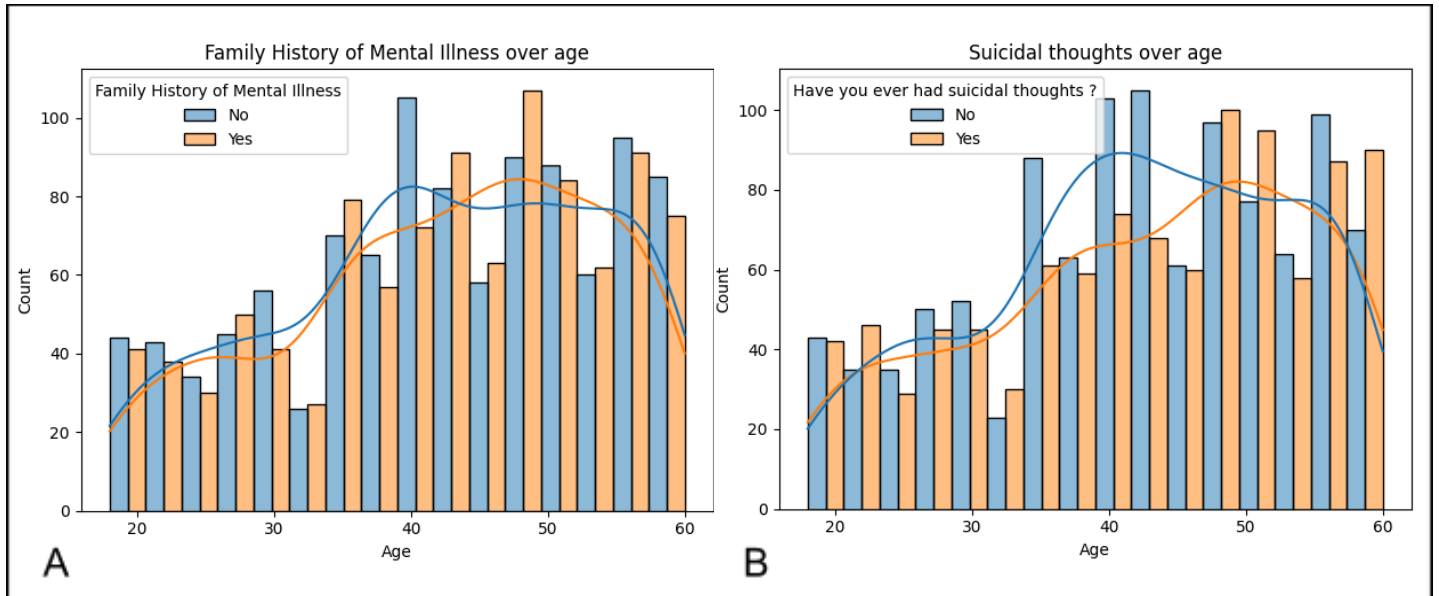


Fig.1.6 two histograms both comparing age to, A)'Family History of Mental Illness, and B)suicidal thoughts.

When observing fig.1.6a the overall distribution is not following a bell-shaped curve or normal-like distribution. Below the age of 35 'No' is a more frequent response than 'Yes' to having a family history of mental illness . From age 35 to 50 the 'Yes' frequency becomes comparable or higher than the 'No' response. After age 50, the distribution becomes more similar, although 'Yes' continues to show several peaks. The Kernel Density Estimate (KDE) peaks later around 47 for 'Yes' and is slightly wider than the 'No' curve that peaks around 40.

Fig.1.6b illustrates participants below age 35 most common response is 'No'. Ages 35 to 45 continue to have greater value of 'No', peaking at age 42. Ages 45 to 60 has a substantial increase in the 'Yes' responses and often exceeds 'No'. The KDE for 'No' peaks sharply around 40, whereas 'Yes' has a broader distribution, starting lower and peaking at age 50.

Due to the data being heavily skewed towards middle aged participants, this could have influenced the age-related trends. Both attributes become more prevalent relative to the 'No' response in older age groups (45-60) compared to younger groups

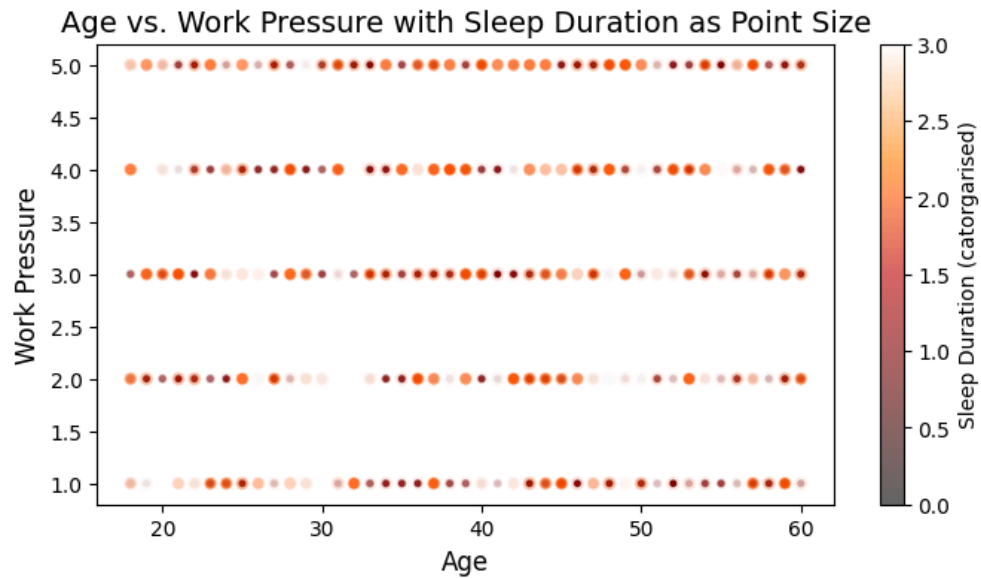


Fig.1.7 Scatter plot showing the relation between Age, Work pressure, and sleep duration.

Fig.1.7 suggests there is weak correlation, that as Work Pressure increases, Sleep Duration decreases. The effect of Work Pressure is clearly evident when comparing a value of 1 and at 5. However, the value of 3 does not follow this trend as many individuals have a very low Sleep Duration.

There is a clear pattern that those who are younger(<32), and those older (>50) get more sleep. There is a clear cluster of darker plots in the center of the graph indicating that middle aged individuals with a work pressure around 3 get the least amount of sleep.

Overall there is a very weak correlation that as age and work pressure increases, sleep duration decreases.

Table.1.5 Sleep Duration encoded values to original categorical values

Column	Sleep Duration
0	<5 hours
1	5-6 hours
2	7-8 hours
3	>8hours

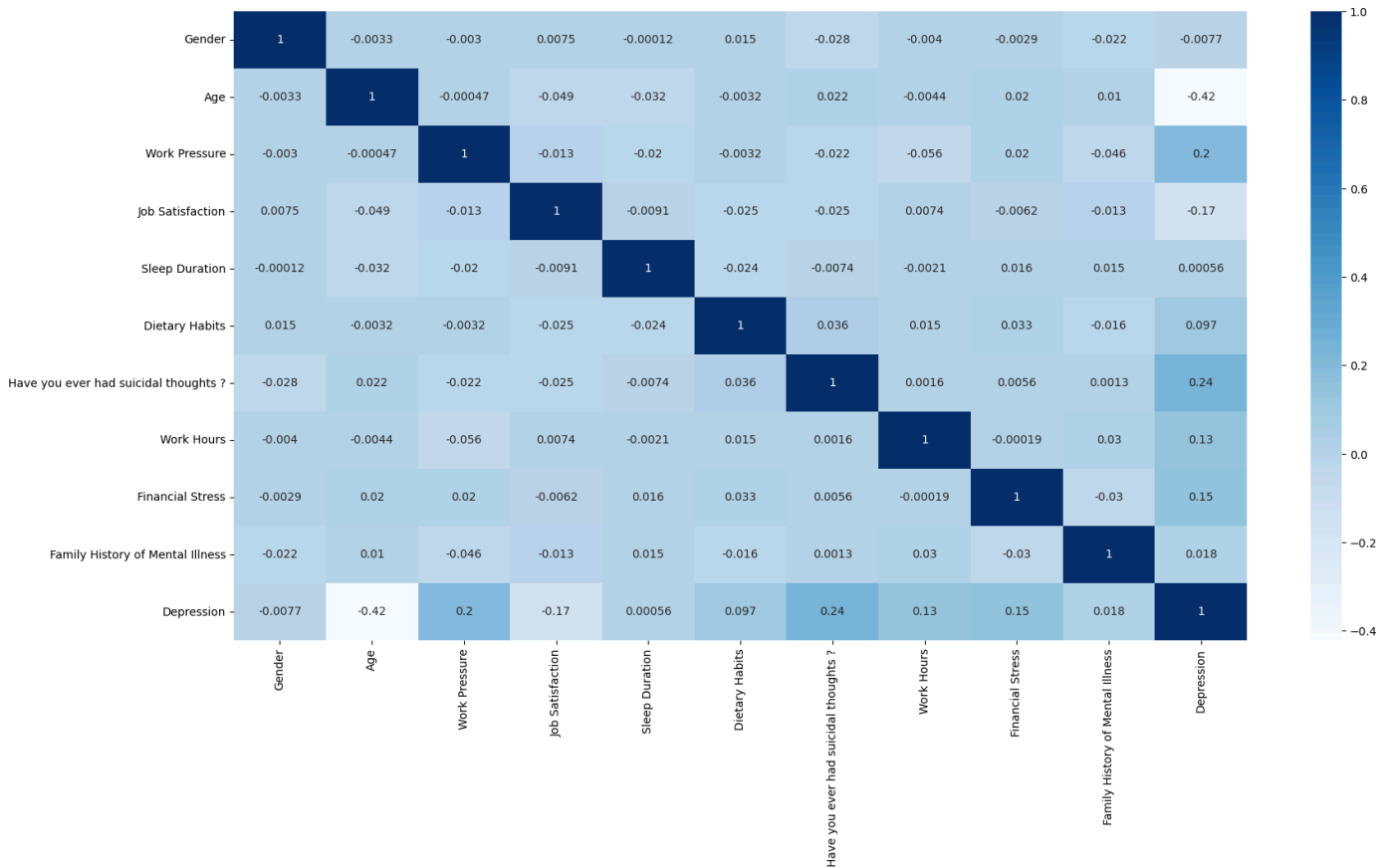


Fig.1.8 Correlation matrix (Heatmap) of all attributes.

Table.1.6 top 4 attributes with the highest correlation in descending order

Correlation (r)	Attribute(s)
Moderate negative (-0.42)	Age and Depression
Weak positive (0.24)	Suicidal Thoughts and Depression
Weak positive (0.2)	Work Pressure and Depression
Weak negative (-0.17)	Job Satisfaction and Depression

This heatmap illustrates that no relationship has a strong correlation and a few attributes show little to no correlation: sleep duration and dietary habits, plus financial stress with all other variables. However it does indicate the independent variables like Gender, Sleep Duration, and Work Hours as the correlation coefficients are consistently close to zero.

The lack of correlation between expected pairings suggests that either: the true relationship are non-linear and not captured by Pearson's r. Or the measured variables are poor proxies for the underlying latent constructs. Or the sample composition is unique/significant confounding variables that mask the expected linear relationship.

2 Classification

2.1 Result Metrics

Due to the medical nature of this prediction, the primary performance metric to base our conclusion on will be the recall score. The dataset has an imbalance seen in Fig 2.1, with only 9.9% of patients suffering from depression. Uneven weight means that the model could have a high accuracy score by predicting “not depressed” for every patient, failing to identify those suffering from depression.

By studying the recall ability of the models,. While the primary metric should be recall, precision should be monitored, preventing the inverse problem where every patient is predicted as “depressed”, supplying an overwhelming amount of false positives.

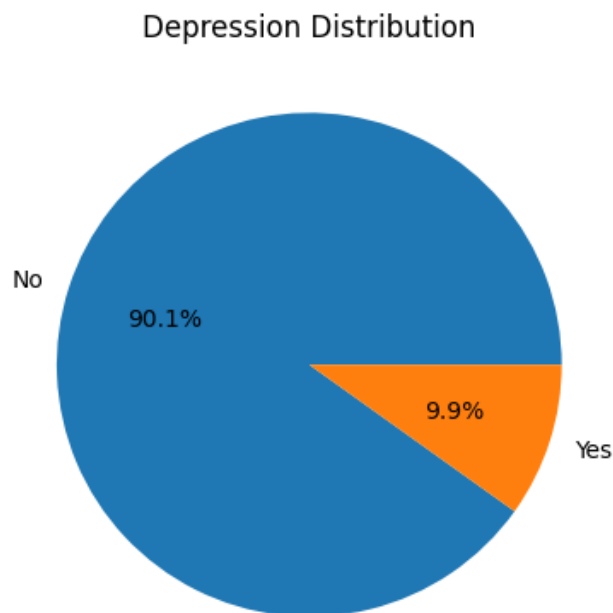


Fig. 2.1 Pie chart showing the distribution of depression

2.2 Choice of Models

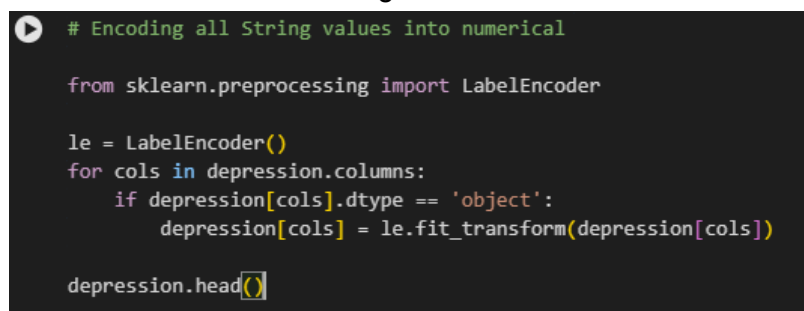
Table 2.1 Model benefits and drawbacks

Decision Tree	• Quick results • Fast to process (though not essential for this task) • It may be more important when working with a larger version of the dataset
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Random Forest	• Robust to noisy data • Can handle an imbalance in the dataset • Won't lose performance when dealing with less relevant attributes
K-Nearest Neighbour	• Effective due to similarity-based prediction • Can struggle with noisy data
Logistic Regression	• Works well with small datasets • Effective with linear patterns (seen in rule mining) • Suitable choice for this task

2.3 Data Preparation

The dataset was first checked for null values; none were found. The next step was ensuring all data was in numerical format so that it could be interpreted by the models correctly. The encoding process can be seen in Fig.2.2. Converting categorical features into numerical values is essential for most machine learning models to work.



```
# Encoding all String values into numerical

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
for cols in depression.columns:
    if depression[cols].dtype == 'object':
        depression[cols] = le.fit_transform(depression[cols])

depression.head()
```

Figure 2.2 Encoding categorical data as

All Features were retained, and feature scaling was applied to ensure all features are treated equally, and no single attribute was overemphasised due to differences in size. Finally, due to the inherent imbalance in the data discussed in [2.1 Result Metrics](#), class weights were applied. This was done so that the models didn't just predict 'not depression' with the goal of getting over 90% accuracy.

2.4 Results

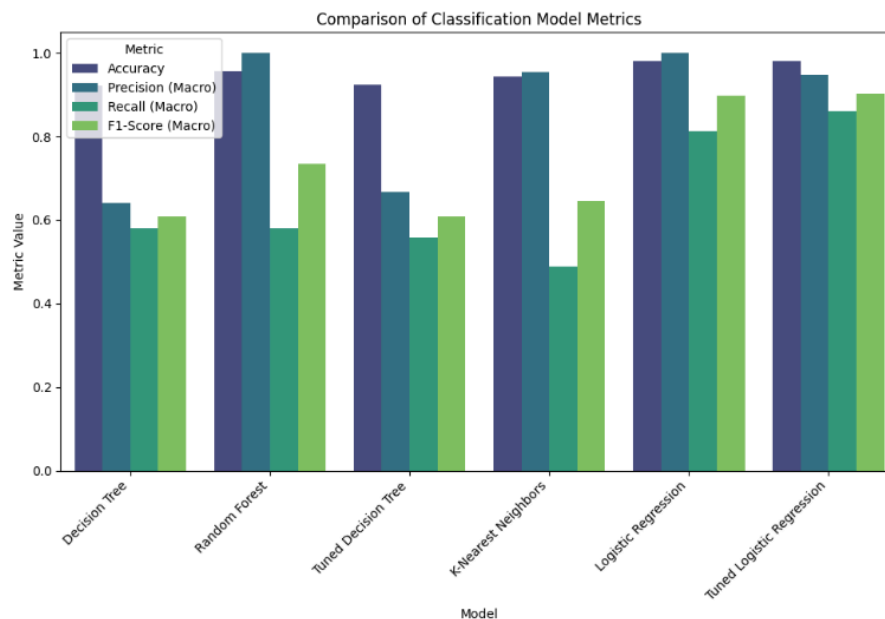


Fig. 2.3 Comparison of classification model metrics

After the class weighting was added, the recall score for each model shot up, with logistic regression reaching a 100% recall score for True Positives. Adding `'classWeight='balanced'` (seen in Fig 2.4) forces the models to achieve a higher recall score by treating both “depressed” and “Not depressed” with the same importance.

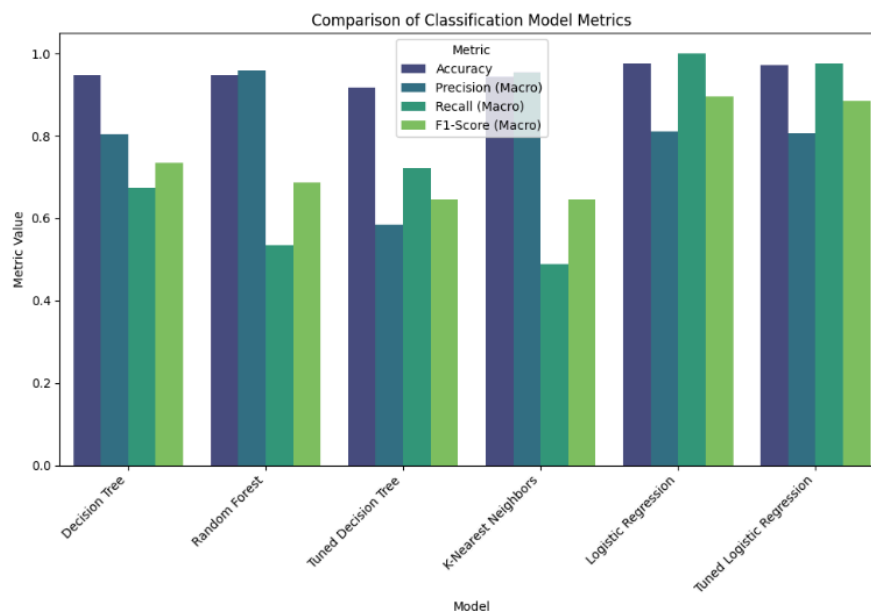


Figure 2.4 Comparison of classification metrics after class weighing was added

Table 2.2 Results for recall and precision of true positives

Model	Decision Tree	Random Forest	Tuned Decision Tree	K-Nearest neighbour	Logistic Regression	Tuned Logistic Regression
Recall score	0.67	0.53	0.74	0.49	1.00	0.98
Precision score	0.78	0.96	0.59	0.95	0.81	0.81

```

LR = LogisticRegression(max_iter=10000, class_weight='balanced')

LR.fit(x_train, y_train)

y_pred = LR.predict(x_test)

accuracy_score(y_test, y_pred)

print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

LR_accuracy = accuracy_score(y_test, y_pred)
LR_f1_score = f1_score(y_test, y_pred)
LR_recall = recall_score(y_test, y_pred)
LR_precision = precision_score(y_test, y_pred)

[[358  10]
 [  0  43]]

```

	precision	recall	f1-score	support
0	1.00	0.97	0.99	368
1	0.81	1.00	0.90	43
accuracy			0.98	411
macro avg	0.91	0.99	0.94	411
weighted avg	0.98	0.98	0.98	411

Fig 2.5 Logistic regression results

2.5 Logistic Regression

The results presented in Fig. 2.2 highlights that logistic regression is the highest performing model. This was further improved with the inclusion of `classWeight='balanced'` parameter, which compensates for the imbalance by assigning more weight to the minority class. As seen in Fig 2.5 the model had a 1.00 recall score for identifying patients with depression; this means every true case of depression was detected. The lower precision shows a tendency to flag individuals as depressed who were not; this trade-off is within the acceptable range given in [2.1 Result Metrics](#) and is somewhat common when maximising recall.

This model was fine-tuned using `GridSearchCV`. Fig. 2.5 shows the results of this parameter search, enabling the evaluation of different parameter combinations. Results

show minimal improvement, suggesting that the default and class-weighted configurations for Logistic Regression were optimal for the dataset.

2.6 Comparison of Low Performing Models

As observed in *Table 2.2* other models performed less than ideally with Random Forest and K-Nearest neighbour suffering from prioritising the majority class during training and Decision Tree most likely suffering from overfitting. The Tuned variation of Decision Tree performed better for recall score however gave far more false positive results.

Although K-Nearest Neighbour and Random Forest had high precision scores as discussed in [2.1 Result Metrics](#) the low recall score shows that there was a preference to predicting the majority class whereas logistic regression treated both classes with equal weighting achieving a high precision and recall score.

3. Regression

3.1 Regression Tree Regressor

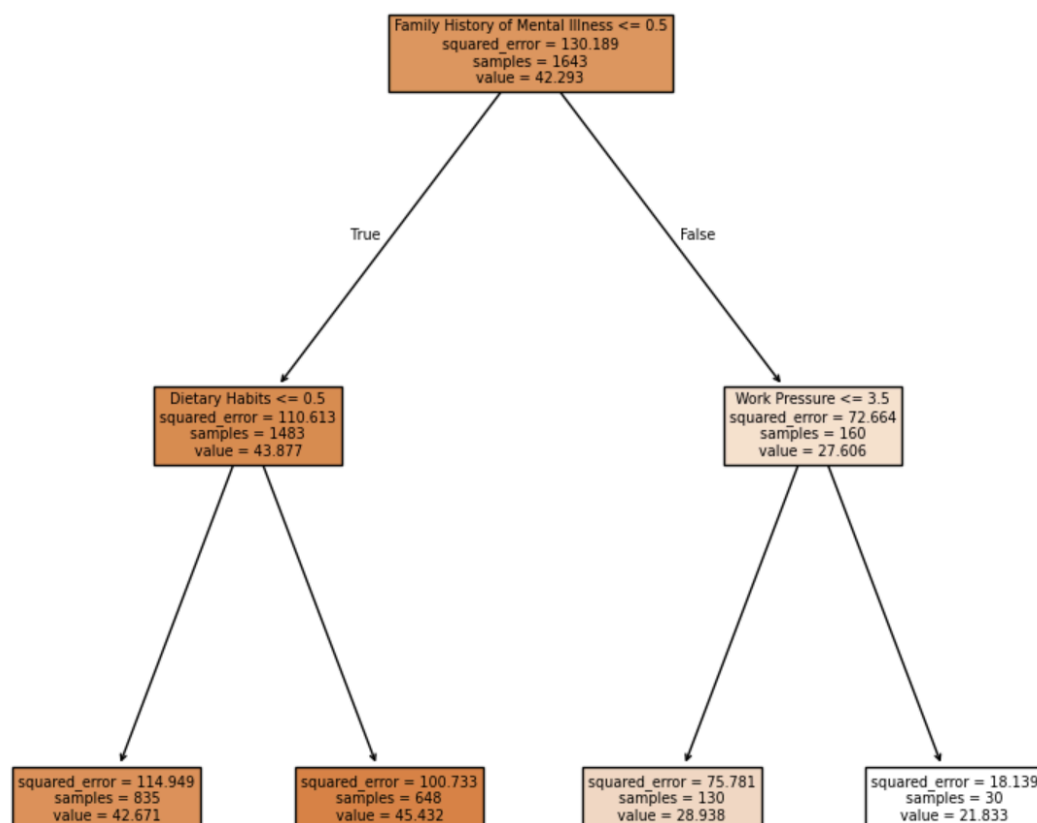


Fig.3.1 Decision Tree Regressor to predict 'Age'

The root node indicates that 'Family History of Mental Illness' is the strongest determinant of the target variable 'Age'. The first split, true branch(Family History ≤ 0.5) further splits on

Dietary Habits ≤ 0.5 , suggests the lifestyle factor diet is influential when family history is low. When false branch (Family History > 0.5) splits on Work Pressure ≤ 3.5 , indicates that occupational stress is relevant when Family History is present.

The root node error (130.189) has a high variance, which decreases down the tree. The leaf node errors range from 18.139(Work Pressure) to 114.949(Dietary Habits). The branch with a small sample size of 30 has the lowest error, indicating high homogeneity.

The left and right branch sample distribution indicates that the model prioritises lifestyle factors (Dietary Habits) for most individuals but the smaller sub group(Work Pressure) has more precise prediction (lower error).

The tree is shallow so there is low variance- less overfitting but potential bias- under fitting if relationships are complex. There is good interpretability but may miss nonlinear interactions beyond these splits.

3.2 Linear Regression

OLS Regression Results						
Dep. Variable:	Age	R-squared:	0.234			
Model:	OLS	Adj. R-squared:	0.230			
Method:	Least Squares	F-statistic:	62.36			
Date:	Sat, 22 Nov 2025	Prob (F-statistic):	9.18e-111			
Time:	17:21:20	Log-Likelihood:	-7650.1			
No. Observations:	2054	AIC:	1.532e+04			
Df Residuals:	2043	BIC:	1.538e+04			
Df Model:	10					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	39.2852	1.104	35.584	0.000	37.120	41.450
Gender	-0.0340	0.444	-0.076	0.939	-0.906	0.838
Work Pressure	0.8701	0.161	5.401	0.000	0.554	1.186
Job Satisfaction	-1.0551	0.159	-6.640	0.000	-1.367	-0.744
Sleep Duration	-0.3144	0.201	-1.565	0.118	-0.708	0.079
Dietary Habits	0.4871	0.271	1.798	0.072	-0.044	1.019
Have you ever had suicidal thoughts ?	3.3235	0.459	7.233	0.000	2.422	4.225
Work Hours	0.2083	0.060	3.495	0.000	0.091	0.325
Financial Stress	0.7774	0.159	4.884	0.000	0.465	1.090
Family History of Mental Illness	0.5605	0.445	1.259	0.208	-0.312	1.433
Depression	-20.2007	0.817	-24.735	0.000	-21.802	-18.599
Omnibus:	92.553	Durbin-Watson:	2.046			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	52.948			
Skew:	-0.240	Prob(JB):	3.18e-12			
Kurtosis:	2.377	Cond. No.	45.0			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

Fig. 3.2 Ordinary Least Squares (OLS) regression analysis

With 'Age' being the dependent variable and $R^2 = 0.234$ plus $\text{Adj. } R^2 = 0.230$ this gives a modest 23% variance in age, suggesting other unobserved factors influence age. F-statistic = 62.36, $p < 0.001$ indicates this model is statistically significant.

Table 3.1 key predictors from OLS(fig.3.2)

Attributes	coef	p> t	prediction
Depression	-20.707	< 0.001	Strong negative association with age; individuals reporting depression tend to be younger.
Suicidal Thoughts	3.373	< 0.001	Positive association; older individuals are more likely to report having suicidal thoughts.
Work Pressure	0.870	< 0.001	Higher work pressure correlates with slightly older age.
Job Satisfaction	-1.051,	< 0.001	Lowest satisfaction is linked to younger age.
Dietary Habits	0.487	0.073	Minimal effect healthier habits may correlate with older age but not strongly significant.
Family History of Mental Illness	0.5605	0.208	Not statistically significant.

Model diagnostics of Omnibus & Jarque-Bera being < 0.001 confirms rejection the null and non-normal residual(errors) which are possible signs of model misspecification, outliers/heavy tails and Possible heteroskedasticity.

Durbin-Watson = 2.046, so no major autocorrelation issue. The skew is mild plus near-normal kurtosis.

3.3 Model Comparison

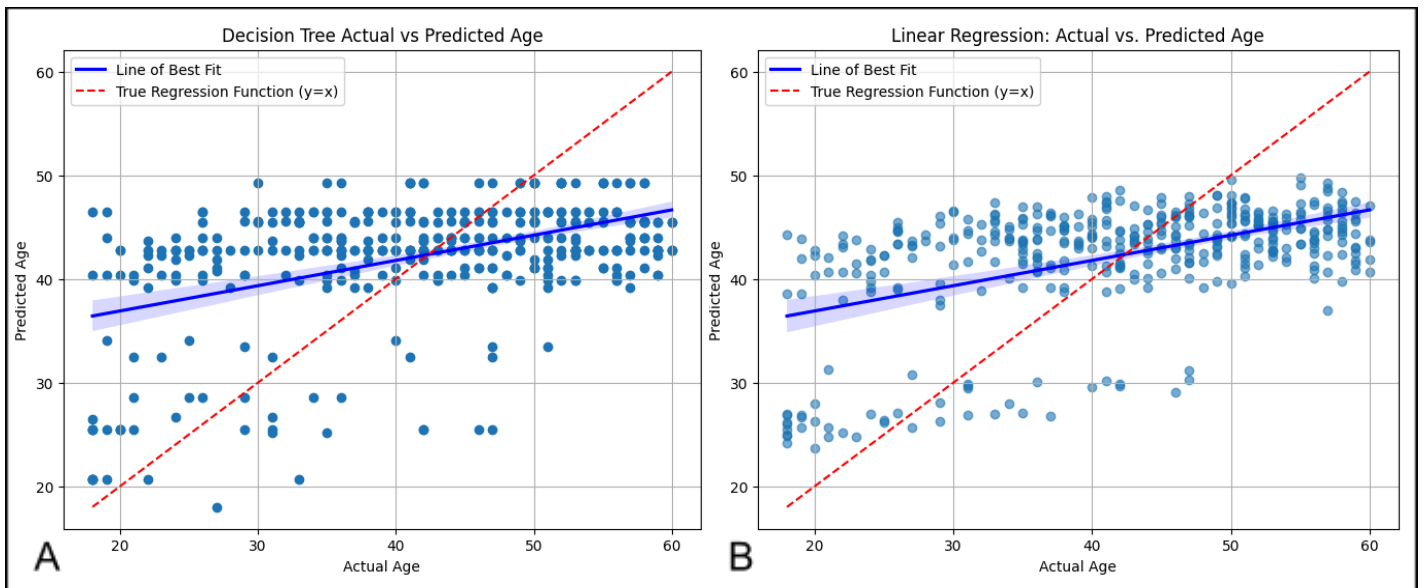


Fig.3.3 Scatter Graphs comparing the actual vs predicted 'Age' outcomes from regression models. A) Decision Tree Regressor outcomes. B) Linear Regression outcomes

Both graphs from Fig.3.3 have a cluster of plots around 40 to 45 near to the interception of the line of best fit and where $y=x$. Values sitting on or closest to the true regression line are found roughly between the ages of 37 and 50 for both models.

The decision tree has a slightly weaker correlation possibly due to this model struggling with extrapolation, and tends to predict near the mean of training data for unseen values. This is typical because they partition data into discrete regions rather than modelling continuous trends.

Fig3.3b shows a slightly stronger relationship between actual and predicted. Linear regression captures the overall trend better, but residual spread suggests underfitting for non-linear patterns.

Table.3.2 Comparison of predictive performance metrics for Decision Tree Regressor and Linear Regression models.

	Regression Tree	Linear Regression
Mean Absolute Error (MAE)	8.681720331905165	8.318267339367466
Mean Squared Error (MSE)	110.11647386209356	101.17400413688972
Root Mean Squared Error (RMSE)	10.49363968611909	10.058528925090872
R2 score	0.18688896930525423	0.25292105805814413
Adjusted R2 score	0.16656119353788557	0.23424408450959777

It is highlighted in table.3.2 that linear regression has a slightly better prediction accuracy when looking at the MAE score. Then looking at MSE and RMSE linear regression has a lower error, confirming more consistent prediction. It also has a higher R^2 and Adjusted R^2 showing more variance in age and generalises better given the number of predictors. Overall, the relatively low R^2 (<0.3), suggests that 'Age' is influenced by additional factors not captured within this dataset.

4. Rule Association Mining

4.1 Data Preparation

Data preparation includes encoding all numerical values into categorical values (Fig. 4.1). This ensures compatibility with the Apriori algorithm. Choosing categorical values will reduce processing time and improve the readability of the resulting rules. The final step is to convert all the data into string values, pass these values into a new data frame

4.2 Rule Mining

For the rule mining process there were two choices for support and one choice for precision. Initially we used a higher support of 0.25 to gather some rules that are more present in the dataset. However, it was observed when analysing rules in *Table 4.1* that all the results were

tending towards “No Depression”. To fix this problem we tuned the support value down to 0.05 to find rules that had “Yes Depression” present, these can be seen in *Table 4.2*.

Table 4.1 Minimum support and confidence

Parameter	Value	Explanation
Minimum Support	0.20	Rules must appear at least 20% of the data set to ensure that irrelevant rules are not displayed
Minimum Confidence	0.50	Ensures that rules generated have sufficient predictive quality. A balance between being too strict on precision allows interesting rules to appear but filters out irrelevant rules.
Adjusted Support	0.05	After observation of rules in <i>Table 4.2</i> the minimum support was lowered to allow attributes such as “Depression Yes” to appear

4.2.1 Filtering

- (1) Lift-based filtering — Only rules with a lift of more than 1 were retained from the original output. A lift of over 1 suggests a positive association between the antecedent and consequent. By filtering out all the rules with lifts below 1, it keeps the data free of noise and removes irrelevant or meaningless rules.

4.3 Results Insight

Table 4.2 ‘Support = 0.5’ rules to analyse

	Items	Antecedent	Consequent	Support	Confidence	Lift
1	{Depression=No, Age={51.75, 60.0}}	{Age={51.75, 60.0}}	{Depression=No}	0.250243	1.0	1.10967
2	{Have you ever had suicidal thoughts ?=No, Family History of Mental Illness=No}	{Family History of Mental Illness=No, Have you ever had suicidal thoughts ?=No}	{Depression=No}	0.257546	0.974217	1.08106
3	{Dietary Habits=Healthy, Depression=No}	{Dietary Habits=Healthy}	{Depression=No}	0.309153	0.932452	1.034715

4	{Dietary Habits=Moderate, Depression=No}	{Dietary Habits=Moderate}	{Depression=No}	0.2926	0.910606	1.010473
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4.3.1 Rule 1

Rule 1, seen in *Table 4.2* shows that individuals between the ages of 51 and 60 will always fall into the no depression category, as demonstrated by the perfect confidence score of 1.0. The support score is irrelevant in this case due to the grouping of numerical values into 4 unique categories (discussed in [4.1 Data preparation](#)). What is clear from this is that certain ages are less susceptible or perhaps less accepting of a depression diagnosis. This rule would be useful to support classification for the depression attribute.

4.3.2 Rule 2

Rule 2 reports that participants who had “no suicidal thoughts” and “no family history of depression” are highly likely to belong in the “No Depression” category. This is reflected by the high confidence score of 97.4%. The support indicates that this combination of antecedents make up 26% of the dataset, giving a fairly weak but positive association to be used when predicting depression.

4.3.3 Rule 3 and 4

These rules go hand in hand of sorts as there are three categories available to answer the “dietary habits” section. The “healthy” and “Moderate” are both present with over 90% confidence, While the lift scores are relatively low there is a clear association with dietary habits and not having depression. “Unhealthy” not being present in the ruleset suggests an inverse association with “unhealthy” having a higher presence of “Yes Depressed” however due to the skew towards “No Depression” there is such little support available to find rules such as these.

4.4 Post Analysis Changes

After analysing the initial results (from table 4.2), the search criteria was amended as a bias highlighted in the dataset meaning that patients who answered “Depression = Yes” would not be retrieved due to the support being too low (below 10%).

Table 4.3 ‘Support = 0.05’ Rules to analyse

	Items	Antecedent	Consequent	Support	Confidence	Lift
1	{'Depression=Yes', 'Have you ever had suicidal thoughts ?=Yes', 'Age=(17.999, 35.0]}'	{'Depression=Yes'}	{'Have you ever had suicidal thoughts ?=Yes', 'Age=(17.999, 35.0]}'	0.06377	0.6453	4.8024

2	{'Depression=Yes', 'Age=(17.999, 35.0]}'	{'Depression=Yes'}	{'Age=(17.999, 35.0]}'	0.07887	0.798	2.88583
3	{'Job Satisfaction=(0.999, 2.0]', 'Depression=Yes', 'Have you ever had suicidal thoughts ?=Yes'}	{'Depression=Yes'}	{'Job Satisfaction=(0.999, 2.0]', 'Have you ever had suicidal thoughts ?=Yes'}	0.053554	0.54187	2.761798

4.4.1 Rule 1 (*Table 4.3*)

Rule 1 as observed in *Table 4.2*) demonstrates a very high association between people diagnosed with depression and people between the ages of 18 and 35 reporting having suicidal thoughts. The support for this rule while at first glance is very low a support of 6.3% this is actually fairly high as only 9.9% of the dataset suffered with depression. This rule also suggests a similar insight to rules discussed in [4.3.1 Rule 1](#) and [4.3.2 Rule 2](#) suggesting that a younger patient reporting suicidal thoughts are to be depressed whereas the rules in *Table 4.2* Discuss the other side of this.

4.4.2 Rule 2 (*Table 4.3*)

Rule 2, demonstrates that “depression yes” was more common amongst participants between the ages of 18 and 35. Whilst the support shows to be low in this rule similar to [4.4.1 Rule 1 \(Table 4.3\)](#) due to “depression yes” containing less than 10% off the dataset the support is irrelevant in this instance. This rule suggests a very similar pattern to that if [4.4.1 Rule 1 \(Table 4.3\)](#) however the lift being 2.8 suggests that there is a positive association between the items.

4.4.3 Rule 3 (*Table 4.3*)

Rule 3 shows a positive association between patients who suffer depression and those who have reported feeling suicidal thoughts (similar to [4.3.1 Rule 1](#)). Interestingly this rule also shows an association between those with low job satisfaction and depressed patients. This could suggest that patients who find work unenjoyable are more likely to be depressed. From this rule we can infer that real world situations and circumstances can have a direct influence on a patient's depressive state.

5. Clustering

5.1 K-Prototypes Clustering

K-Prototypes clustering is designed to handle mixed data types (numerical and categorical data) simultaneously. It uses a different distance metric that can calculate dissimilarity between both continuous and discrete attributes.

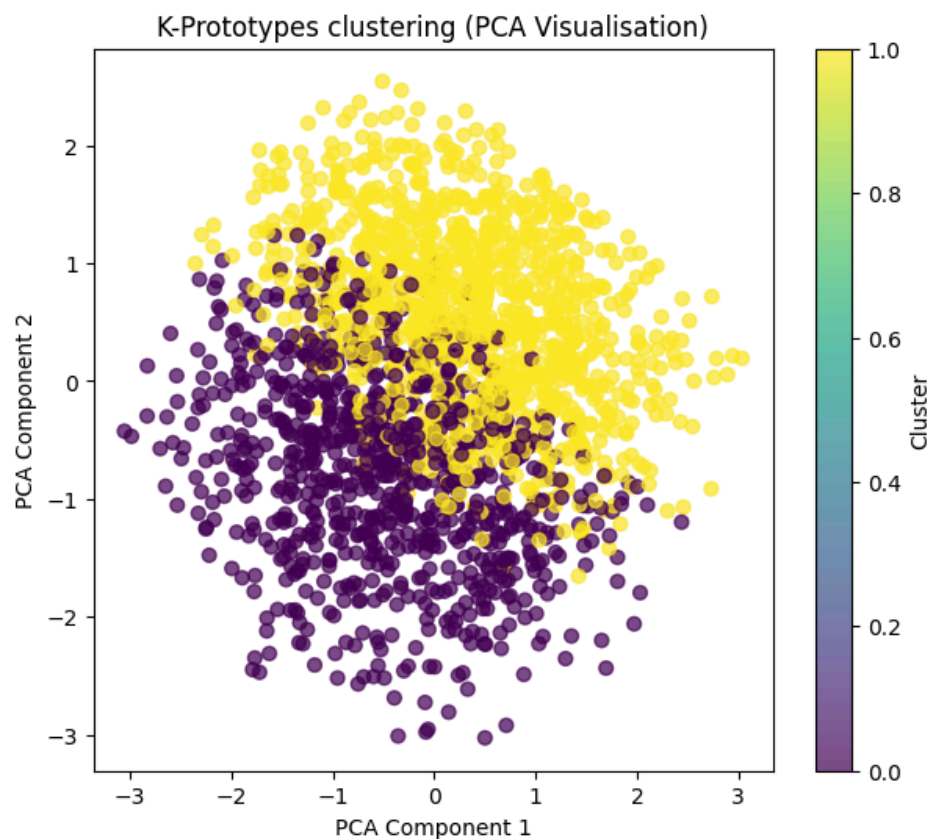


Fig. 5.1 PCA visualisation for K-Prototypes clustering

The plot on Fig.5.1 shows two different coloured clusters. While there is visible separation along the 'PCA component 2' y-axis, the clusters overlap significantly near the center, highlighting moderate cluster distinctiveness.

As K-Prototypes are designed for mixed data, the separation is likely due to the differences in categorical attributes combined with numeric features. PCA compresses this complexity into two components which may obscure some structure.

The overlap indicates that the dataset contains heterogeneous groups with shared characteristics, making perfect segmentation difficult. This could indicate: high intra-cluster variability, possible need for more clusters ($k > 2$) or alternative distance metrics, or dimensionality reduction may have lost some discriminative power.

5.3 HDBSCAN Clustering

HDBSCAN clustering is a powerful, non-parametric technique that advances beyond traditional density-based methods by constructing a cluster hierarchy to identify structures of varying density. Unlike k-means, it automatically discovers the optimal number of clusters and designates low-density regions as noise (outliers). Its strength lies in its ability to extract the most stable and prominent clusters from the hierarchy, ensuring robustness against fluctuating local densities and removing the need for a single, global density threshold (E). Therefore this model is well-suited for high-dimensional /complex datasets where cluster definitions are not uniform.

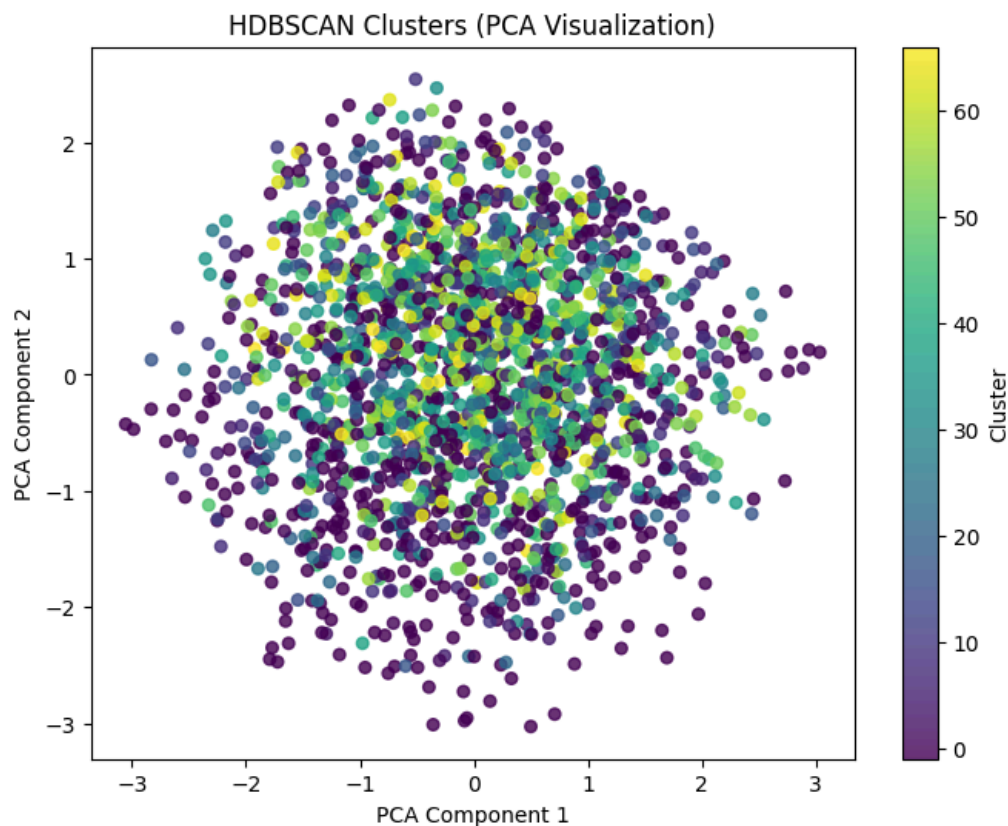


Fig. 5.2 PCA visualisation for HDBSCAN clustering

HDBSCAN has identified many small clusters. This reflects its density-based nature, which does not force all points into clusters (some may be treated as noise).

In PCA space the clusters are highly overlapping, forming a dense core, suggesting the data lacks strong global separation, but this algorithm is capturing local density variations rather than global partitions. The large frequency of clusters and overlaying indicates: data is complex and possibly high-dimensional with subtle patterns, and PCA reduction may obscure meaningful density differences.

5.3 Clustering Algorithm Comparison

Table.5.1 Davies-Bouldin Index for each clustering algorithm.

Algorithm	Davies-Bouldin Index
K-Prototypes	3.11
HBDSCAN	7.01

When comparing Davies-Bouldin Index from each clustering algorithm in Table.5.1, K-Prototype is much lower indicating better separation and compact clustering, However this is still relatively high suggesting clusters are not as distinct or well-separated as they could be. Implying that the within-cluster scatter is relatively high compared to the between-cluster separation.

So even though there is a considerable difference between both algorithms there is still room for improvement and further investigation.

Appendix A

Contribution table:

ID	Contribution	Tasks
Up2110125	50%	Section 2, Section 4, section 5
Up2102739	50%	Section 1, section 3, section 5